LEARNER GUIDE

Negative and fractional indices

Number Systems and Calculator Judgement · Lesson 4. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

A power ladder lists the powers of a base: $2^3 = 8$, $2^2 = 4$, $2^1 = 2$. Each rung down divides by the base. Carry on down and you reach $2^0 = 1$, then $2^{-1} = 1/2$ and $2^{-2} = 1/4$. A negative index gives **the reciprocal**, never a negative number. Between the whole rungs are in-between rungs: two half-steps make × 9, so each half-step is × 3, and $9^{½} = 3 = \sqrt{9}$. Then you prove, with the index laws, that these are the only meanings that keep the laws true.

What you are learning to notice

- Going down one rung divides by the base. So $a^0 = 1$ and $a^{-n} = 1/a^n$ (for any a except 0).
- A negative index never makes the answer negative: $5^{-2} = 1/25$, not –25 or –10.
- The reciprocal of a fraction: the numerator and denominator change places. $(2/3)^{-1} = 3/2$.
- A power of ½ is the square root, and a power of 1/3 is the cube root. For $a^{m/n}$, take the nth root, then the power m.
- The proofs: $a^3 \div a^3$ is both a^0 and 1; $a^{½} \times a^{½} = a$, so $a^{½} = \sqrt{a}$.

Moving through it, one step at a time

For a negative index, work out the positive power, then take the reciprocal. For a fractional index, take the root (the denominator of the index) first, then the power (the numerator). If you are unsure, sketch the power ladder and count the steps from 1.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — Down the ladder, and why

Work out 2^0 and 2^{-3}, then prove that $a^0 = 1$.

1. $2^2 = 4$, $2^1 = 2$. Down one rung: ÷ 2.	$2^0 = 2 \div 2 = 1$
2. Keep dividing by 2.	$2^{-1} = 1/2$, $2^{-2} = 1/4$
3. One more rung.	$2^{-3} = 1/8 = 1/2^3$
4. Proof: by the law, $a^3 \div a^3 = a^0$. But a number divided by itself is 1.	$a^0 = 1$ ($a \neq 0$)

$2^0 = 1$ and $2^{-3} = 1/8$: the reciprocal of 2^3.

Example 2 — Sideways on the ladder

Work out $8^{2/3}$ and $8^{-2/3}$.

1. Three third-steps make one whole step, × 8. $? \times ? \times ? = 8$
2. $2 \times 2 \times 2 = 8$, so each third-step is × 2. $8^{1/3} = \sqrt[3]{8} = 2$
3. Two third-steps up from 1: × 2, × 2. $8^{2/3} = (\sqrt[3]{8})^2 = 4$
4. A negative index: the reciprocal. $8^{-2/3} = 1/4$

$8^{2/3} = 4$ and $8^{-2/3} = 1/4$. The root first, then the power, then (if negative) the reciprocal.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (the index laws for whole-number powers, and squares, cubes and their roots), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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